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# Dimensions and barriers for digital (in)equity and digital divide: a systematic integrative review

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## Abstract

**Purpose** – This integrative review was conducted to provide an overview of existing research on digital (in) equity and the digital divide in developed countries.

**Design/methodology/approach** – We searched academic and grey literature to identify relevant papers. From 8464 academic articles and 183 grey literature, after two levels of screening, 31 articles and 54 documents were selected, respectively. A thematic analysis was conducted following the steps suggested by Braun and Clarke and results were reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.

**Findings** – The results showed that most articles and papers were either from Europe or North America. Studies used a range of research methods, including quantitative, qualitative and mixed methods. The results demonstrated four major dimensions of the digital divide among various vulnerable groups, including digital literacy, affordability, equity-deserving group-sensitive content and availability or access to infrastructure. Among vulnerable groups, low-income people were reported in the majority of the studies followed by older adults, racial and ethnic minorities, newcomers/new immigrants and refugees, Indigenous groups, people with disabilities and women. Most reported barriers included lack of access to the internet, digital skills, language barriers and internet costs.

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**Originality/value** – To the best of our knowledge, there have been limited attempts to thoroughly review the literature to better understand the emerging dimensions of digital equity and the digital divide, identifying major vulnerable populations and their unique barriers and challenges. This review demonstrated that understanding intersectional characteristics (age, gender, disability, race, ethnicity, Indigenous identity and immigration status) and their interconnections is crucial for analyzing the dynamics of digital (in)equity and divide.

**Keywords** Digital divide, (In)equity, Vulnerable groups, Barriers, Developed countries

**Paper type** Literature review

## Introduction

With the continued advancement of information and communication technologies (ICT) over the last three decades, it has become hard for people to think of a day when they did not rely on these essential tools (Organization for Economic Co-operation and Development, 2019). Digital technology benefits us in multiple ways by lowering the barriers to accessing education, commerce, e-participation, health care, social connectedness and entertainment (Chayko, 2014; Haight, Quan-Haase, & Corbett, 2014). The widespread use of digital technology was instrumental in providing various social and health services during isolation and quarantine orders brought on by the COVID-19 (De', Pandey, & Pal, 2020). This global crisis saw exponential growth in digital technology usage and online accessibility for shopping, telework and distance learning (Beaunoyer, Dupéré, & Guitton, 2020; Guitton, 2020) making the critical societal function of digital technologies all the more obvious as they spread throughout the world (Saeed, Bader, Al-Naffouri, & Alouini, 2020).

Amid the worldwide wave of digitalization, equitable access and usage of ICT has emerged as a crucial aspect in the society (Robinson, Chen, Schulz, & Khilnani, 2018). Although there is widespread ICT usage, a digital divide has emerged that can be tied to a person's access to or ability to use these technologies (Vargo, Zhu, Benwell, & Yan, 2021). While ICT has brought enormous benefits to society, many argue that it has further exacerbated pre-existing disparities and is expanding the digital divide. Similarly, existing inequities stemming from historical injustice and systematic exclusion can be exacerbated due to existing digital divide, which resulting in increased issues of inclusions and exclusions (Hadziristic, 2017) both digital and otherwise. This creates socioeconomic disadvantages among people that intersect with the gradient of social stratification based on socioeconomic factors [such as age, gender, ethnicity, race, educational level and income, among others] (Hadziristic, 2017; Lu, Yu, Liu, & Yao, 2003). The issue of the digital divide not only raises concerns about digital equity (Yates, Kirby, & Lockley, 2015) but also has scholars arguing that ICT access and training related to digital literacy are fundamental human rights required to access various essential day to day services, including critical healthcare, education and social services (Murray, 2021; The city of Cassey, 2021; United Nations Human Settlements Programme, 2021). Even though equity of getting digital services is a global concern (Baum, Newman, & Biedrzycki, 2014), developed countries need to work on bridging the uneven digital services distribution internally and worldwide to fully see the benefits of digitalization (Smith, St Amour, & O'Halloran, 2018).

The COVID-19 pandemic, as a global crisis, has demonstrated the need for careful understanding and exploration of the dimensions and context of the digital divide in a holistic way to unravel the factors and barriers at play. The early policy rhetoric and discursive framing related to digital divide were mainly based on accessibility issues, especially access to devices and peripherals (Deursen, Dijk, & A.G.M, 2019), which is also considered a structural approach to conceptualizing the digital divide (Servaes & Oyedemi, 2016). However, this dominant framing is severely criticized (Gorski, 2009; Mäkinen, 2006) and has sparked scholarly debates among many researchers as it tends to overlook other important dimensions of digital divides such as disparity of skills [substantial information skills –

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operationalizing] or effective navigation skills in digital world and use the internet [which is often considered as the second-level digital divide] (Deursen, Alexander, & van Dijk, 2014). Given the growing worldwide digital divide, which became exacerbated by the COVID-19 pandemic, research with a focus on reviewing the various dimensions of digital divides among vulnerable groups is needed. This can be grounded in the access dimension and first-level divide, with area wide range of systematic reviews, focused on identifying the digital divide levels, determinants of ICT skills and outcomes and/or digital divide at the organizational level (Lythreatis, Singh, & El-Kassar, 2022; Scheerder, van Deursen, & van Dijk, 2017). When we move away from the popular narrative of physical/material access to digital devices as the sole means of understanding the digital divide, many argue that this concept is, fundamentally, an equity concern (Ahmed, 2020; Albert, Flournoy, & LeBrasseur, 2009). This has caused the issue of digital inequity to gain considerable attention in the policy debates with redressing digital equity becoming a priority issue for promoting inclusion, diversity and the overall well-being of the citizens/city dwellers.

To the best of our knowledge, there is a limited attempt to thoroughly review the literature on digital equity and the digital divide to better understand the emerging dimensions, context, the major vulnerable populations and the unique barriers/challenges they face. Using the methodological approach suggested by Whittemore and Knafl (2005), this integrative review synthesized knowledge on digital equity and the digital divide in both academic and non-academic research activities (Whittemore & Knafl, 2005). Relevant literature on digital in(equity) and the digital divide was systematically reviewed with the following specific objectives: (1) to understand the context of the digital divide/gap (2) to identify the major dimensions of the digital divide and which groups are most vulnerable. The findings of this review will assist policymakers in taking necessary steps to improve digital equity in developed countries like Canada.

## Materials and methods

### *Community-engaged research approach*

We followed a community-engaged research approach to promote the production and sharing of knowledge (Burke *et al.*, 2013; Turin, Lasker, Rahman, Rumana, & Chowdhury, 2023), which gives priority to equitable relationships and ensures that our partners have the power to make decisions at every level of the research process (Turin, Chowdhury, Haque *et al.*, 2021). Our strategy entails actively interacting with communities, creating inclusive cooperation and advocating knowledge sharing. As part of our outreach activities, we engaged and collaborated with the team member of the City of Calgary to conduct this research on digital inequity (Turin *et al.*, 2022). Together with The City team, we conducted this systematic review and synthesized knowledge from existing research on digital equity and the digital divide.

### *Integrative review*

We conducted a systematic integrative review of existing knowledge related to digital (in) equity and divide following Whittemore and Knafl's methodological approach (2005). This integrative review approach allows generating knowledge from both empirical and theoretical sources. The steps of integrative literature review as is followed by this study are: (1) problem identification, (2) identifying relevant studies, (3) study selection, (4) data extraction, (5) data synthesis and presentation of results (Whittemore & Knafl, 2005). The Preferred Reporting Items for Systematic Review and Meta-Analysis [PRISMA] guidelines was followed for the reporting of the steps associated with this review (Page *et al.*, 2021) (Appendix Table 1).

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*Stage 1: problem identification.* The problem of this study has been identified in collaboration between the researchers and The City of Calgary representatives. Digital inequity is a critical issue in policy debates to promote inclusion, diversity and the well-being of urban people. However, to the best of our knowledge, there is a dearth of literature reviews on addressing digital equity, barriers forming the digital divide and the impact disadvantaged groups in the community encounter. Therefore, to understand the evolving dimensions and context of the digital divide, we identified the following research questions:

*RQ1.* What are the key determinants of digital (in)equity and digital divide in developed countries?

*RQ2.* Which groups are particularly at risk of digital (in)equity and the digital divide, and what are the barriers frequently reported within these groups?

*Stage 2: identifying relevant studies.* We systematically searched academic and grey literature through various databases to identify relevant studies. Major academic databases included MEDLINE, Scopus and Web of Science and grey literature included Google Scholar, ProQuest and OAISter (see Table 1 for full list). Our search term included both keywords and MeSH terms, which included three key themes: "digital (in)equity", "developed countries" and "(in) equity" (Table 2). The Boolean operators "OR" and "AND" were used to connect the

| Academic articles | Grey literature |
|---|---|
| • Web of Science<br>• Scopus<br>• Academic Search Complete<br>• Canadian Research Index<br>• MEDLINE<br>• SocIndex with FullText<br>• Communication & Mass Media Complete<br>• IEEE Xplore digital library: Standards | • Google Scholar<br>• ProQuest (theses and dissertations)<br>• OAISter (WorldCat)<br>• National Digital Inclusion Alliance<br>• Canadian Radio-television and Telecommunications Commission<br>*Internet Scan*:<br>• Yahoo<br>• Google<br>• Bingo |

**Source(s):** Authors' own work

**Table 1.** Academic and grey literature databases searched

| |
|---|
| *Keywords for* digital (in)equity |
| (Digital* OR "digital literacy" OR "information technology" OR "digital technology" OR technology OR internet OR "information and communications technology" OR ICT OR computer OR mobile OR phone OR smartphone OR "smart devices" OR cyber OR web OR "data literacy" OR "information literacy") AND |
| *Keywords for* developed countries |
| ("OECD countr*" OR "developed countr*" OR "Western countr*" OR "Organisation for Economic Co-operation and Development" OR "developed nation*" OR "advanced countr*" OR "advanced nation*" OR "industrialized nation*" OR "industrialized countr*" OR "high-income countr*" OR "first world count*" OR "MEDC countr*" OR "More economically developed countr*") AND |
| *Keywords for (in)equity* |
| (Equity OR inequity OR divide OR inclusion OR exclusion OR gap OR inequality OR apartheid OR equality OR disadvantage* OR inconvenien* OR access* OR unfair OR fair OR justice OR injustice OR discrimination OR bias OR unjust OR need* OR barrier* OR obstacle* OR limitation* OR deficit OR shortage OR inadequate OR poverty OR scarcity OR insufficient OR scant) |

**Source(s):** Authors' own work

**Table 2.** Search terms and search strategy

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synonymous and different keywords, respectively. In addition, as part of the grey literature search, we conducted an internet scan (Turin, Chowdhury, Ekpekurede *et al.*, 2021) using Google, Bing and Yahoo! search engines to find non-academic documents about digital (in) equity and divide. To identify further studies or reports, we reviewed the reference list of all the selected academic and non-academic sources.

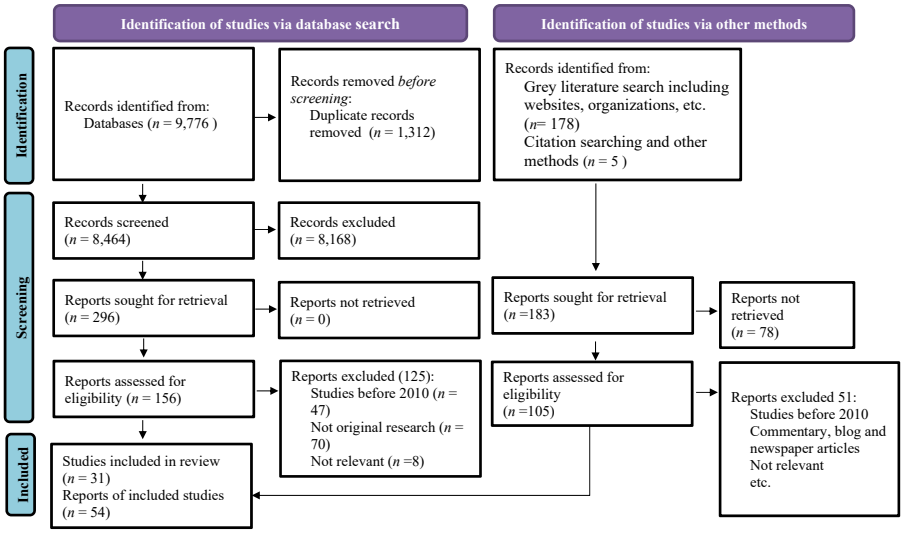
*Stage 3: study selection.* The "Population-Concept-Context (PCC)" framework was followed to set the screening criteria to include relevant studies (Gumberg Library, 2024). The inclusion criteria for identifying studies are presented in Table 3. A two-stage screening process was employed in this study, including title-abstract and full-text. Following systematic searches of all selected databases, we imported all articles into COVIDENCE to start the screening after duplicates were automatically removed. Two independent reviewers carefully screened the titles and abstracts in the COVIDENCE. Any conflicts were resolved through discussion with a third reviewer. The same process was followed in both stages of screening as well as while evaluating grey literature. A PRISMA flow diagram (see Figure 1) was used for tracking study number at each stage of the review (Page *et al.*, 2021).

*Stage 4: data extraction.* Two trained reviewers independently collected pertinent data from each selected paper. Data were gathered regarding the citation, study objective(s), sample characteristics (such as target demographic, population size and age) and study type. We also extracted data on who is more vulnerable to the digital divide and their most reported

| Inclusion criteria | Exclusion criteria |
|---|---|
| 1. Original research paper | 1. Studies not in English |
| 2. Developed countries | 2. Rural areas |
| 3. Urban area | 3. Published before 2010 |
| 4. English | |

**Source(s):** Authors' own work

**Table 3.** Inclusion and exclusion criteria




**Source(s):** Figure by the authors; Adapted from diagram for systematic review (Page *et al.*, 2021)

**Figure 1.** Flow diagram of the search and selection process for the systematic integrative review

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barriers. Finally, data on the determinants of digital in(equity) and digital divide were extracted. We charted the data in Microsoft Excel [Microsoft Corporation, Redmond, Washington, USA]. It might be difficult to assess the quality of primary research in integrated reviews that contain a range of study designs from different empirical sources (Whittemore & Knafl, 2005). Whittemore and Knafl recommend assessing the quality of studies that report outlier findings to determine if such findings may be caused by methodological inconsistency. Although we found two papers that gave us outlier findings during our data analysis, we did not find them due to methodological inconsistency. We included different methodological papers on a variety of outcomes when reporting.

*Stage 5: data analysis and reporting the results.* In the final stage, data were compiled to find codes and develop themes. We followed Braun and Clarke's framework for thematic analysis, including (1) familiarizing with the data, (2) generating initial codes, (3) searching for themes, (4) reviewing themes, (5) defining and naming themes and (6) writing the report (Braun & Clarke, 2006). Data were scrutinized and compared iteratively by the research team to generate key themes and sub-themes. The results have been organized based on those themes and interpreted and contextualized. This was possible due to our community-engaged approach that allowed representatives from The City and the researchers to contemplate the themes, reflect and interpret according to their perspectives.

## Results

Our initial search found 9,776 citations in the scholarly databases. We imported all citations to COVIDENCE and 8,464 articles remained after removing the duplicates. In the title and abstract screening stage, we removed 8,168 articles. The remaining 296 documents were chosen for full-text review. In the full-text review stage, we finally selected 31 relevant articles for data extraction. In addition, 183 records were found in the grey literature scan. Of these, 78 records were removed during the titles and abstracts screening stage. The remaining 105 documents went through full-text review. Finally, 54 documents from the grey literature search met the inclusion criteria (Figure 1).

### *Study and sample characteristics*

The details of the study characteristics are provided in Appendix Table 2 and Appendix Table 3. Appendix Table 2 depicts the study characteristics identified through scholarly database searches. The majority of the studies were conducted in the European Union countries (n = 14), followed by the North American countries (n = 5). A range of research methods were used across studies. Majority of the studies used quantitative research methods (n = 26) followed by qualitative methods (n = 3) and mixed-methods research (n = 2). Appendix Table 3 presents the study characteristics of the selected reports identified from grey literature. Most of the documents were reports with some dissertations, case studies and policy briefs, which predominantly were from the USA and Canada followed by New Zealand, Sweden, Australia and the UK These documents focused on a diverse population, including older population, general population country-level data, people who are unhoused, young population, female domestic workers and residents of retirement homes.

### *Emerging dimensions of the digital divide*

There has been a major shift in the definition of the dimensions of the digital divide from the homogenization of access barriers to a more holistic account which considers other socio-cultural dynamics. Table 4 highlights the key dimensions of the digital divide among various vulnerable groups and the wide variety of factors shaping these dimensions. These factors have been identified and categorized into four major themes (Figure 2).

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| Dimensions | Sub-categories | Barriers | Study (N) | Source documents* |
|---|---|---|---|---|
| Digital literacy/ knowledge | Technological Literacy | Individuals' limited proficiencies to effectively, safely, and responsibly usage, and management of ICT | 7 | Foteinou (2010), Heponiemi *et al.* (2021), Huang and Chen (2010), Kim Andrew Eugni and Jeong Mi-Kyeng (2010), Lamberti *et al.* (2021), Nam, (2010), Neves *et al.* (2013) |
| | Language proficiency | Accessing or utilizing digital content in English language may be challenging for non-English speaking people | 2 | Boesch (2012), Chang *et al.* (2016) |
| | Awareness of available digital services | A limited knowledge of available digital services involves issues about how to find and access the desired content and services | 2 | Chang *et al.* (2016), Heponiemi *et al.* (2020) |
| | Information navigation literacy | A lack of information and navigation skills prevents people from accessing and understanding IT hardware and software, which in turn, hindering them from achieving their desired services | 3 | Heponiemi *et al.* (2020) Lamberti *et al.* (2021), Lytras and Şerban (2020) |
| | Safe use of the internet | A lack of internet safety is also a concern as digital technology becomes more widespread, where people are becoming increasingly aware of their digital security and well-being | 2 | Heponiemi *et al.* (2021), Shaffer (2011) |
| Affordability (costs) | Internet Plan | • A lack of access to affordable home internet and data plans is a significant barrier that worsens the digital divide<br>• Relative cost of the internet and its's associated value (total data allowance and maximum speed) are important factors that shape the choices of plans and usage patterns | 7 | Deursen *et al.* (2019), Dilmaghani (2018), Ee (2013), Hartnett (2017), Heponiemi *et al.* (2020), Kim Andrew Eugni and Jeong Mi-Kyeng (2010), Lamberti *et al.* (2021) |
| | Device | The cost of digital devices (mobile phones, laptops, computers, and tablets) also a barrier to access ICT and related services | 6 | Deursen *et al.* (2019), Dilmaghani (2018), Hartnett (2017), Heponiemi *et al.* (2020), Lamberti *et al.* (2021), Nielsen *et al.* (2018) |
| | Connection cost | Connection costs (such as modems and routers) and the cost of internet data package, broadband internet connection and internet data package subscriptions also barriers for many low-income households | 5 | Dilmaghani (2018), Hartnett (2017), Heponiemi *et al.* (2020), Lamberti *et al.* (2021), Nielsen *et al.* (2018) |

(*continued*)

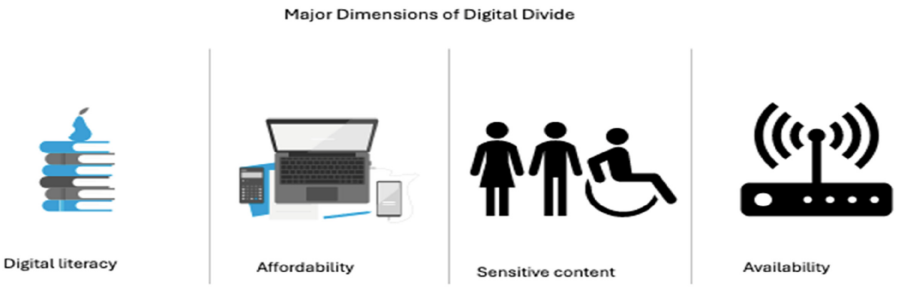
**Table 4.** Major dimensions and barriers of digital in(equity)

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| Dimensions | Sub-categories | Barriers | Study (N) | Source documents* |
|---|---|---|---|---|
| Equity deserving group sensitive content and services | | The lack of culturally appropriate content, activities, programs, and services is critical barriers to achieving digital equity for equity-seeking groups | 3 | Neves *et al.* (2013), Tirado-Morueta *et al.* (2021), White *et al.* (2011) |
| Availability/Access (infra-structure) | Availability of internet | A lack of availability of affordable broadband or Wi-Fi connections hinder people to enjoy the benefits of the digital world | 5 | Dilmaghani (2018), Hartnett (2017), Heponiemi *et al.* (2021), Lamberti *et al.* (2021), Nielsen *et al.* (2018) |
| | Quality of internet | The lack of quality and speed in broadband and Wi-Fi connections, as well as slow upload and download speeds, are also crucial barriers to bringing disadvantaged people into the digital world | 4 | Hartnett (2017), Heponiemi *et al.* (2021), Nielsen *et al.* (2018), Sanz and Turlea (2012) |

**Note(s):** * Details of these sources are available in Appendix Table 2 and Appendix Table 3
**Source(s):** Authors' own synthesis

**Table 4.**




**Source(s):** Figure by the authors

**Figure 2.**
Dimensions of digital (in)equity and divide

The results show that individuals' digital literacy is critical factor in the digital divide. Digital skills and literacy shape people's digital experience, which can help their ability to overcome the fear of using electronic devices or navigate the digital space creatively, effectively and safely. Similarly, information navigation literacy helps people access the internet and develop the necessary skills for safe internet usage. While awareness of existing digital content is critical for gaining access to the desired digital services, the lack of plain language content may lead to challenges for people whose first language is not English when attempting to utilize services in English. This connects to the importance of culturally appropriate desired and sensitive content and services for equity-seeking populations, such as the elderly and new immigrants. This is highlighted in many studies as a critical factor in encouraging these groups to participate equally in the digital world. In terms of affordability, a lack of access to affordable devices and internet plans significantly exacerbates the digital divide between low and high-socioeconomic groups. Being that low-income individuals often cannot afford the cost of reliable speed internet services; the accessibility and availability of quality internet remain crucial elements in bridging the digital divide.

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*Vulnerable groups and key reported barriers to digital equity*

Our results find that digital gap remains a persistent issue in developed countries with differentiated compounding effects on various population groups across the socio-economic spectrum, demographic range and geographical scale. Table 5 summarizes the major vulnerable populations most likely impacted by the digital divide and the main barriers they are facing. According to the reviewed studies, pre-existing structural inequalities, socioeconomic status and intersectional identities shape the vulnerability and put them at risk of experiencing the digital divide. This study identified population groups, as more likely to be vulnerable to digital exclusion and systematic marginalization due to their unique exposure to single or multiple barriers. These groups include low-income individuals, older adults, indigenous groups, racial and ethnic minorities, newcomers and refugees, people with disabilities and women.

The reviewed studies uncovered that income is the most reported vulnerability due to the wide variety of factors influencing people with low-income access to ICT, including the high monthly expenses of internet subscription plans and the high price of devices. This group commonly lacks access to laptops and desktop computers, as well as digital abilities, which prevents them from regularly using online platforms which, when coupled with the rate of technology changes, serves to widen the digital gap. The second most reported vulnerable group is older adults. This group tends to lack internet access due to barriers such as a lack of pertinent digital tools, digital skills and motivations required to learn to navigate digital platforms properly and with confidence. This can be further compounded for people of racial-ethnic minorities, new immigrants and refugees who experience major obstacles in language, technological literacy and the cost of the internet and devices. Other compounding barriers may be experienced by people with mobility issues and/or special needs, using the internet and online communities can be an additional layer of struggle due to various issues, such as a lack of accessible content and/or access to devices required for using online platforms. Examining barriers within Indigenous communities reveals that the absence of internet access and digital literacy poses substantial challenges, which hinder their access to health, education and social connectivity. The findings also show that women are susceptible to the digital divide and exclusion due to the addition of safety concerns, and online harassment challenges placed on top of internet access, affordability challenges and a lack of technological proficiency.

Moreover, two different studies indicate single parents and members of the 2SLGBTQ+ community are vulnerable to digital (in)equity. The findings reveal concerning issues related to online safety and limited internet access among individuals identifying as 2SLGBTQ+ (The city of Cassey, 2021). The 2SLGBTQ group faces challenges in using and navigating the digital world, which may happen because of the negative representation in the media and prejudice against the 2SLGBTQ Community. Similarly, single parents experience digital exclusion, primarily due to the high costs associated with broadband internet packages, exacerbating barriers to essential digital services and information (Andrey, Masoodi, Malli, & Dorkenoo, 2021). These results emphasize the pressing need for additional research to shed light on the specific challenges faced by these communities in navigating digital technology.

## Discussion

The synthesis of the current literature on digital (in)equity and the digital divide highlights the significance of digital equity for vulnerable groups. This review reveals that both institutional and personal factors contribute to digital inequity as a multi-dimensional problem. The findings emphasize the essential role of possessing digital skills for accessing online services and critically evaluating online content. Individuals lacking digital literacy may not fully realize the advantages of technology and be reluctant to engage in online platforms.

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| Population sub-groups | Most reported barriers | Study (N) | Source documents * |
|---|---|---|---|
| Low-income people | Limited internet access and digital skills | 25 | Bailey and Nyabola (2021), Boesch (2012), Breed (2019), Carson *et al.* (2021), Chang *et al.* (2016), Cherewka (2020), Forman *et al.* (2020), Good Things Foundation (2021), Hartnett (2017), Heponiemi *et al.* (2020), Horrigan (2021), House and Urban Development (2016), Islam (2021), Martinovic and Freiman (2018), Moore (2021), Mubarak *et al.* (2020), Murray (2021), Nielsen *et al.* (2018), Sanz and Turlea (2012), Schmidt (2005), Social Planning Council of Ottawa (2021), The city of Cassey (2021), United Nations Human Settlements Programme (2021), Vogels (2021), Zwimpfer *et al.* (2017) |
| Older adults | Lack of technological skills, digital Literacy, fear, and safety concern | 17 | Andrey *et al.* (2021), Bucea (2018) Cherewka (2020) Good Things Foundation (2021), Islam (2021), Kim and Jeong (2010), Lamberti *et al.* (2021), Martinovic and Freiman (2018), Moore (2021) Nam (2010), National Science Foundation (2016), Puretz and Aguinaldo (2021), Schmidt (2005), Social Planning Council of Ottawa (2021), The city of Cassey (2021), Vazquez *et al.* (2015), Zwimpfer *et al.* (2017) |
| Racial and ethnic minorities | Limited internet access | 8 | Andrey *et al.* (2021), Carson *et al.* (2021), Cherewka (2020), Dilmaghani (2018), Moore (2021), Schmidt (2005), United Nations Human Settlements Programme (2021), Zwimpfer *et al.* (2017) |
| Newcomers/New immigrants and refugees | Limited internet accessibility | 7 | Bailey and Nyabola (2021), Carson *et al.* (2021), Cherewka (2020), Dilmaghani (2018), Good Things Foundation (2021), United Nations Human Settlements Programme (2021), Zwimpfer *et al.* (2017) |
| Indigenous groups | Limited internet access and digital skills | 6 | Good Things Foundation (2021), Murray (2021), Resta and Laferriere (2015), The city of Cassey (2021), United Nations Human Settlements Programme (2021), Zwimpfer *et al.* (2017) |
| Persons with disabilities | Internet cost, language barriers, digital skills, and lack of easy-to-understand content | 5 | Good Things Foundation (2021), Puretz and Aguinaldo (2021), The city of Cassey (2021), United Nations Human Settlements Programme (2021), Zwimpfer *et al.* (2017) |
| Women | Technological knowledge, Access, Internet Safety | 4 | Carson *et al.* (2021), Foteinou (2010), Good Things Foundation (2021), The city of Cassey (2021) |

**Note(s):** * Details of these sources are available in Appendix Table 2 and Appendix Table 3
**Source(s):** Authors' own synthesis

**Table 5.**
Most vulnerable groups (who experience digital inequity) and key barriers reported by them

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This study also highlights factors beyond individual control, such as a lack of accessibility and affordability, and a dearth of relevant content for equity-deserving groups, as key determinants of digital (in)equity. These factors significantly impact individuals' ability to acquire essential digital skills, enhance personal digital experience and – most importantly – effectively use ICT for tangible outcomes. While discussions have moved away from physical/material access to capacity building, motivation, usage and outcomes some studies emphasize digital divide as a series of divides (Barzilai-Nahon, 2006) as well as disparities (Dimaggio, Hargittai, Celeste, & Shafer, 2004). Resonating with these arguments our findings suggest that personal and/or institutional factors may contribute to the digital divide among vulnerable people, separately or collectively.

Our reviewed studies indicate that individuals from low-income groups lack ICT access and high-quality broadband services due to their disadvantaged socio-economic conditions (Mubarak *et al.*, 2020; Nielsen *et al.*, 2018). In Canada, most low-income people lack equitable access to affordable broadband internet connections (Andrey *et al.*, 2021), making internet cost a primary barrier to digital exclusion (Social Planning Council of Ottawa, 2021). Also, inadequate access to devices like computers or laptops poses a significant barrier to online services and digital skills development (Forman, Basma, & Gourley, 2020). For marginalized individuals without proper devices and reliable internet, navigating the digital world becomes challenging, which hinders their acquisition of transferable digital skills. Given the emerging dynamics of the digital divide and uneven internet diffusion, Mäkinen (2006) emphasized addressing the digital gaps in the opportunity to utilize the technologies in inclusive ways that not only empower people but also create equitable participation space. Therefore, we need efforts to provide more affordable broadband and devices for effective participation and essential digital literacy. Specialized programs and free online services are also suggested to reduce the digital divide for low-income individuals.

Among vulnerable groups identified in this review, older adults primarily are at risk of digital exclusion due to a lack of internet access and digital literacy skills (Andrey *et al.*, 2021; Deursen *et al.*, 2019; Wilco, 2021). Exploring further, research indicates that older adults are found to be less interested in using the Internet and learning new skills (Eugni & Mi-Kyeng, 2010). This lack of enthusiasm may stem from cumulative factors such as a lack of relevant and culturally sensitive digital content, perceived technology usability challenges and difficulties in learning new digital skills (Nam, 2010; Eugni & Mi-Kyeng, 2010). Older adults also may be less interested in using the internet due to their physical and mental health conditions because older people have a lack of confidence resulting from a lack of experience with online safety, security, privacy and disinformation (Mubarak *et al.*, 2020; Murray, 2021; Zwimpfer *et al.*, 2017).

Similarly, people with disabilities face a higher risk of digital exclusion and vulnerability due to limited content accessible to those with physical health-related issues. They may need special arrangements for learning, cognition, hearing, vision and physical movement and may also experience a lack of confidence in using technology. This common experience is rooted in their limited access to assistive technologies and easy-to-understand digital content and websites (The city of Cassey, 2021; United Nations Human Settlements Programme, 2021; Zwimpfer *et al.*, 2017).

Socioeconomic factors and intersectional characteristics including gender, race and ethnicity, Indigenous identity and immigrant and refugee status also increase the risk of digital inequity and the digital divide. While affordability and limited internet access are common barriers for all vulnerable people leading to difficulty achieving important inclusion in social interaction, health, education and other benefits from online services, an incomplete digital experience could negatively affect people's motivation for the digital world and limit their capacity to use digital services. For example, factors such as language obstacles and a

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dearth of culturally relevant information and services result in an incomplete digital experience which may prevent Indigenous communities, racial and ethnic minorities, newcomer and refugee groups and other groups from being motivated to use digital spaces. The demotivational and negative attitudes can act as barriers to digital inclusion and building digital skills. Our findings indicate that members of certain races and ethnicities [e.g. Black, Hispanic and foreign-born] are more likely to have less education and/or less digital literacy (Office of the Innovation and Technology, 2022). However, the challenges faced by Indigenous groups in accessing the internet may differ from those encountered by other vulnerable populations. Many Indigenous communities, particularly those residing in reservations or remote areas, grapple with inadequate infrastructure. Insufficient roadways, a lack of electricity and an absence of reliable internet connectivity compound the challenges faced by these communities (Smillie, 2005).

Furthermore, women may have greater chances of experiencing exclusion from the digital world due to some unique challenges, including an unsafe online environment and abusive comments. Moreover, due to structural barriers, such as e-infrastructure and discrimination, women are also lagging compared to men (Foteinou, 2010; United Nations Human Settlements Programme, 2021), which may jeopardize women's access to digital services. This may happen due to the limited opportunities for women compared to men to acquire useful ICT literacy and skills.

This systematic integrative review examines digital (in)equity and the digital divide, focusing primarily on developed countries while we need to acknowledge that these issues also affect developing nations. (Deursen *et al.*, 2019; Shairgojri, 2022). Research from India, Brazil and Kenya reveals diverse and complex issues of digital (in)equity and divide among vulnerable populations (e.g. across gender, socio-economic class, etc.). A study in India pointed out a significant digital divide between rural and urban areas where rural people have limited skills, infrastructure and investments to access to high-quality internet service (Shairgojri, 2022). The gendered dimension of the digital divide in India presents another significant challenge, highlighted in a recent report by GSM Association, which stated that women in India are 41% less likely than men to use mobile internet (GSM Association, 2021). A study in Brazil identified insufficient infrastructure as a major barrier to the first level of the digital divide, where older people, the poor and underserved populations face the greatest challenges (Nakayama *et al.*, 2023). Research in Kenya showed disparities exists among people having digital devices and skills, and that digital divide is a common problem driven by factors related to infrastructures, high internet cost and having low or no digital literacy to use technologies successfully (Okello, 2024). Digital (in)equity and the digital divide are pervasive issues worldwide, disproportionately affecting vulnerable subgroups in the society.

### *Strengths and limitations*

Our knowledge synthesis project has several strengths. One is its comprehensive search strategy, encompassing academic databases and grey literature. Another key strength of this review is a diverse research team, including The City of Calgary personnel and multidisciplinary researchers. To ensure comprehensive perspectives, all team members contributed to shaping each step of the review from search and screening to analysis and writing the manuscript. Nevertheless, this review has certain limitations. Digital (in)equity is a complex and multifaceted topic that can be understood in multiple ways. For example, there may be distinct challenges associated with digital equity in medicine and education. Although our research seeks to incorporate insights from a variety of disciplines, it may not fully cover all the factors relevant to each area. Furthermore, the quality of the selected studies was not assessed due to the inclusion of diverse study types.

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*Policy implications*

This integrative review provides an in-depth understanding of the various dimensions of digital inequity, vulnerable groups, their unique challenges and most importantly, highlights the need for appropriate actions and policy interventions to eliminate digital inequity and divide for all. As illustrated in the study, the digital divide is a multi-faceted problem stemming from a variety of factors. Eliminating digital inequity and reducing the divide requires continued collaboration and participation from all levels of government – federal, state/provincial and municipal – as well as civil society and non-governmental organizations. For example, to address the issue of access barriers and enhance digital equity among underserved rural communities, there is a need for increased governmental investment in high-quality internet connections to reach those remote areas. It also includes investment in programs that focus on enhancing basic digital skills and competencies for effectively using technologies, irrespective of education or socio-economic conditions. Policy considerations for the internet safety of women and other minorities, including older adults, ethnic minorities, newcomers, refugees, people with disabilities and Indigenous groups, are needed to address their unique challenges. Policy interventions could include creating culturally sensitive digital information and content for all. Content created in this way will increase the relevancy of digital resources as well as the accessibility of services to these vulnerable groups.

## Conclusion

This review highlighted challenges and barriers contributing to the multifaceted problems of digital (in)equity. Understanding and addressing the problem of digital (in)equity from multiple viewpoints has become a popular academic and practical trend. The term digital divide is used to describe both the issue of ICT accessibility as well as the underlying socioeconomic and cultural factors that influence ICT adoption and usage. It emphasizes differences in people's needs, perspectives and abilities, as well as differences in skill and literacy levels. The digital divide is a complex phenomenon with overarching issues which require a holistic understanding of the context of the divide and multi-level and multi-scalar collaboration. This review argued that understanding the intersectional characters, (including age, gender, disability, race-ethnicity, Indigenous identity, immigration status, etc.) and their interconnection, is key to exploring the barriers and critically analyzing the emerging dynamics of digital inequity.

The modern era of digitalization has opened enormous opportunities while creating systematic exclusion across intersectional groups with differentiated impacts. The results suggest that prioritizing state investment in cost-effective broadband connection, targeted programs and training and culturally sensitive content in different languages should be initiated to increase affordability and access to the Internet for vulnerable and equity-deserving groups. Studies recommended culturally appropriate online and offline training programs, human-centered digital services, community-led digital literacy programs and raising public knowledge of the available digital learning resources to increase vulnerable groups' digital literacy and capability. The studies also recommended developing a digital equity plan through coordinated efforts at the federal, state and local levels to guarantee that everyone has the required digital resources and access. This systematic review, therefore, suggests taking context-specific measures to better comprehend the root causes of the digital divide rather than explicitly focusing on the symptoms, with an active collaboration among governmental, civil society, non-governmental organizations and community.

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## Appendix

The Appendix for this article can be found online.

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